

Jackson Ferguson

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Education

University of Victoria

Victoria, BC

B.Sc. Major Combined Physics and Astronomy

Graduated Spring 2026

High Performance Coursework: Quantum Computing (100%), Stellar Astrophysics (99%), Probability & Statistics (97%), Computer Assisted Physics (91%), Experimental Physics (88%), Thermodynamics (86%).

Technical Skills

Languages:	Python (3.12+), Bash, TOML, YAML, L ^A T _E X
Software Architecture:	CLI Design, AST Parsing (tomlkit), Terminal UIs (rich), Regex
Data & Computation:	Pandas, NumPy, SciPy, Matplotlib, Astropy, Jupyter Lab
DevOps & Build:	Git/GitHub Actions, Docker, uv, CI/CD & Release Automation, Package Distribution
Testing & QA:	Pytest (with pytest-mock), Mypy (Strict Static Typing), Ruff, Hypothesis

Engineering & Research Projects

Protostar: Modular Environment Orchestration Engine *Python, AST Parsing, CI/CD*

- Declarative Architecture:** Architected a strictly decoupled state-machine where independent modules aggregate requirements into a centralized EnvironmentManifest. Enforced a "manifest-first, side-effects-last" paradigm.
- AST Manipulation & Safety:** Engineered a non-destructive configuration engine utilizing AST parsing to safely deep-merge complex dependencies into existing pyproject.toml files, preserving user comments and structural integrity.
- Software Rigor:** Maintained rigorous engineering standards including 100% static type coverage, highly mocked isolated test suites, and strict filesystem sandboxing (path jails, Zip/Tar Slip prevention) for remote payload execution.
- Actionable Telemetry:** Designed a comprehensive diagnostic engine that maps domain exceptions to standardized POSIX exit codes for CI/CD automation, and intercepts systemic crashes to auto-generate encoded issue payloads.

Git Pulsar: Distributed State Daemon *Python, Git Internal Plumbing (DAG), Concurrency*

- Distributed Reconciliation:** Architected a fault-tolerant background daemon that tracks non-linear development streams across multiple physical machines, utilizing Octopus Squash merges to reconcile distributed state drift.
- Concurrency & Atomicity:** Engineered crash-resilient local registries usingfcntl POSIX advisory locking and atomic inode swaps (os.replace + os.fsync) to guarantee zero data corruption during sudden power loss or SIGKILL.
- Chaos Engineering:** Designed a rigorous 3-tier testing matrix encompassing property-based fuzzing (Hypothesis), isolated distributed networking sandboxes, and automated Linux VM provisioning (Multipass) for OS-level field testing.

Star Ground: Deterministic Data Pipeline Engine *Python, pdfplumber, Pytest, Docker*

- Data Normalization Pipeline:** Engineered a unidirectional pipeline with a custom spatial parser and recursive SI-unit parser to translate unstructured PDF tables into strictly validated TypedDict schema states.
- Invariant Testing:** Implemented an aggressive CI/CD matrix using property-based fuzzing and golden-master snapshot regression to guarantee mathematical consistency across legacy documents.

Automated CI/CD Operations Engine *GitHub Actions, Python Automation*

- Pipeline Architecture:** Architected a dedicated infrastructure repository housing reusable GitHub Actions workflows, completely decoupling complex CI/CD and deployment logic from application codebases.
- Release Automation:** Developed standalone Python automation scripts utilizing AST parsing (tomlkit) and PyPI API polling to synchronize semantic versions and automatically deploy Homebrew tap updates.

Galaxy Cluster Dark Matter Analysis (ACO 2670) *Astrophysics Research Project*

- Pipeline Development:** Constructed a data pipeline to clean spectroscopic data for 98+ galaxies, utilizing redshift-space distortion analysis to isolate cluster members from the background field.
- Statistical Modeling:** Applied the Virial Theorem and Gaussian Mixture Models (GMM) to calculate a mass-to-light ratio of $(291 \pm 60) M_{\odot}/L_{\odot}$, providing kinematic evidence of Dark Matter dominance.

Work & Leadership Experience

Co-Lead Event Organizer, Unplugged Events, Victoria, BC *Jan 2025 – Present*

- Managed technical production, signal flow, and venue logistics for live events (100+ attendees).

Server, Stanley Park Brewing, Vancouver, BC *May 2025 – August 2025*

Alpine Ski Instructor, Whistler Blackcomb, Whistler, BC *Nov 2018 – Feb 2020*

Freelance 3D Modeler, Twisted Time Studios, Remote *May 2015 – Aug 2015*

- Designed and created main character models and weapon systems for Extinction: Archaic Genesis.